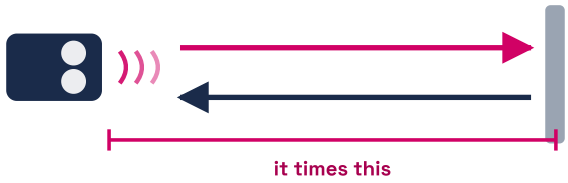
KEEP THIS ONE

Two ways to see without touching

Neither one reaches out and feels anything. Both send something out and wait to see what comes back. **What they send, and what they do with the answer, is the only difference.**

Ultrasonic

It shouts and listens



IT SENDS OUT

SOUND — a click too high for you to hear

IT MEASURES

TIME — how long the echo took to come back

YOU GET

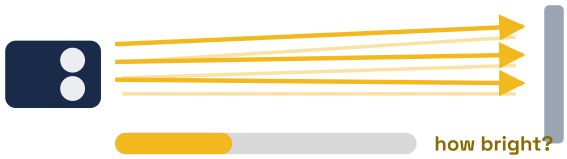
real centimeters, from 3 to 250

IT IS BEATEN BY

soft things, and things at an **angle**

Infrared

It shines and looks



IT SENDS OUT

LIGHT — infrared, just past red, same as a TV remote

IT MEASURES

BRIGHTNESS — how much of it came back

YOU GET

0 to 100. Not centimeters. Out to about 50–70 cm

IT IS BEATEN BY

dark things, **shiny** things, and **sunlight**

THE WHOLE DIFFERENCE, IN ONE LINE

One times a journey. The other judges a brightness.

AND THE RULE UNDERNEATH BOTH OF THEM

Every sensor needs **one** thing from whatever it is looking at. Sound needs something **hard**. Light needs something that **reflects**. Anything without that is not badly seen — **it is not seen at all.**

That is why a cushion is invisible to one and obvious to the other. Not because one is better. **Their blind spots do not overlap.**